IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS 2023

MATHEMATICS 1B

Wednesday 3 May 14:30

Time allowed: 2:00 hours

This exam has four questions, worth 25 Marks each. Answer all questions.

NO CALCULATORS ALLOWED.

Any special instructions for invigilators and information for candidates are on page 1.

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THE QUESTIONS

1. a) Evaluate the double integral

$$I = \int\int_R f(x,y)\, dy\, dx\,,$$

where $f(x,y) = yx^3 + xy^3$, and R is the region bounded by the lines $x = 0$ and $y = x$, and the curves $y^2 - x^2 = 2$ and $xy = 2$. To do so, begin by
(i) sketching the region R;
(ii) next obtain the values of the new variables $u = y^2 - x^2$ and $v = xy$ on the boundaries of R;
(iii) then show that the change of variables gives an integral with constant limits in the $uv-$plane;
(iv) finally, evaluate the integral. [9]

b) A periodic function $f(x)$ with period T, which satisfies the Dirichlet conditions, can be written as a complex Fourier Series:

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{i2\pi nx/T}\,.$$

(i) Show that if k is an integer, then $\int_{-1}^{1} e^{ik\pi x}\, dx = \begin{cases} 0 & \text{for } k \neq 0, \\ 2 & \text{for } k = 0. \end{cases}$ [3]

(ii) Using (i), show that when $T = 2$, the complex Fourier coefficients are

$$c_n = \frac{1}{2}\int_{-1}^{1} f(x) e^{-in\pi x}\, dx$$

for every integer n. [5]

c) A function is defined as $f(t) = t - 1$ for $0 \leq t < 1$. Obtain the complex Fourier series of the odd, period-2 extension of $f(t)$. [8]

2. a) Calculate the Fourier transform $F(\omega)$ of the function [7]

$$f(t) = \begin{cases} \cos(\pi t) & \text{for } |t| \leq 1 \ , \\ 0 & \text{otherwise} \ . \end{cases}$$

Give careful consideration to values of ω where $F(\omega)$ has a zero denominator.

b) We are given a function of two variables $f(x,y) = (x+y)(x^2+y^2-1)$.

(i) Obtain the gradient and the Hessian determinant of $f(x,y)$ and use them to find and classify its stationary points. [6]

(ii) Draw the zero contour, including the stationary points. Sketch the contour plot, explaining your reasoning. [5]

c) Find the function $g(y)$ which makes the following first-order ordinary differential equation (ODE) exact:

$$(x\sin y + g(y) + xe^{xy})\frac{dy}{dx} + ye^{xy} + 2x + g(y) = 0 \ .$$

Hence obtain the solution of the ODE which satisfies $y(1) = 0$. [7]

3. a) Show that the Laplace transform of the third derivative of a function is as follows: [5]

$$\mathscr{L}[f'''(t)] = s^3\overline{f}(s) - s^2 f(0) - sf'(0) - f''(0)\,.$$

b) Use (a) and Laplace transforms to show that the solution of the third-order ODE

$$\frac{d^3x}{dt^3} + \frac{dx}{dt} = \begin{cases} 1 & \text{for} \quad 0 \le t < \pi\,, \\ 0 & \text{for} \quad \pi \le t\,, \end{cases}$$

which satisfies the initial conditions $x(0) = 1\,, x'(0) = 1\,, x''(0) = -1\,,$ is

$$x(t) = \begin{cases} x_1(t) = t + \cos t & \text{for} \quad 0 \le t < \pi\,, \\ x_2(t) & \text{for} \quad \pi \le t\,, \end{cases}$$

giving the correct form of $x_2(t)$. [10]

c) Given the matrices

$$A = \begin{pmatrix} 3 & -3 & 2 \\ -1 & 3 & -1 \\ -2 & \alpha & -2 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 4 & 1 & 3 \\ 3 & -2 & 4 \\ -2 & 3 & -2 \end{pmatrix}$$

(i) Find the value of α for which A^{-1} does not exist, [3]
(ii) If $\alpha = 4$ find $\det(A)$, $\det(B)$, and $\det(AB)$, [3]
(iii) If $\alpha = 7$, use Gauss-Jordan elimination to calculate A^{-1}. [4]

4. a) You are given the matrix $A = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 4 \\ 2 & 5 & 7 \end{pmatrix}$.

(i) Use Gaussian elimination to show why we cannot write $A = LU$, where L is lower-triangular and U upper-triangular. Explain your reasoning. [5]

(ii) Your answer to (i) should have included a single row exchange. Write down the permutation matrix P, with the property that PA can be written as the product of a lower-triangular matrix L, and an upper-triangular matrix U. Now use Gaussian elimination to find U.

With no further calculation, write down the two elimination matrices involved, and hence write down L. [5]

b) Given a 4×3 matrix A, what can we say about rank, nullspace, column space and solutions to $A\underline{\mathbf{x}} = \underline{\mathbf{b}}$, without any calculations? [5]

c) Given

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \\ 10 & 11 & 12 \end{pmatrix},$$

obtain the reduced echelon form of A. State rank, and free and pivot columns and variables. Hence find bases for the four fundamental subspaces of A: nullspace, column space, row space, and left nullspace. In each case, state the dimension and the parent space, as well as a geometric description, if possible. [10]